

```

1 // You can also use an enum...
2 #define DO_SAMPLE 0
3 #define BUF_FULL 1
4 #define BUF_FREE 2
5
6
7 int stm=BUF_FREE;
8 int N=64;
9
10
11 ISR(TIMER2_OVF_vect) {
12     Reload timer
13     Handle 2 kHz output (rising and falling edges)
14
15     if(rising edge){
16         PORTpin = HIGH;
17         ...
18         if(stm==BUF_FREE) stm=DO_SAMPLE;
19         ...
20     }
21     if (stm==DO_SAMPLE){
22         ADCVal[idx]= read a sample;
23         if(++idx==N){
24             idx=0;
25             stm=BUF_FULL;
26         }
27     }
28 }
29
30 void loop() {
31     if(stm==BUF_FULL){
32         make DFT
33         Filter results (magnitude and phase)
34         Display magnitude and phase
35         stm=BUF_FREE;
36     }
37 }

```